

RAFFLES INSTITUTION
2021 Year 6 Preliminary Examination

Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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BIOLOGY

Paper 2 Structured Questions

9744/02
22nd September 2021
2 hours

Candidates answer on Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name in the spaces at the top of this page.

Write in a dark blue or black pen.

You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use

1	/ 10
2	/ 9
3	/ 11
4	/ 12
5	/ 14
6	/ 11
7	/ 14
8	/ 9
9	/ 10
Total	/ 100

This document consists of **28** printed pages.



Raffles Institution
Internal Examination

Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 represents the molecular structure of a transmembrane receptor on the cell surface membrane. It is made up of 485 amino acid residues.

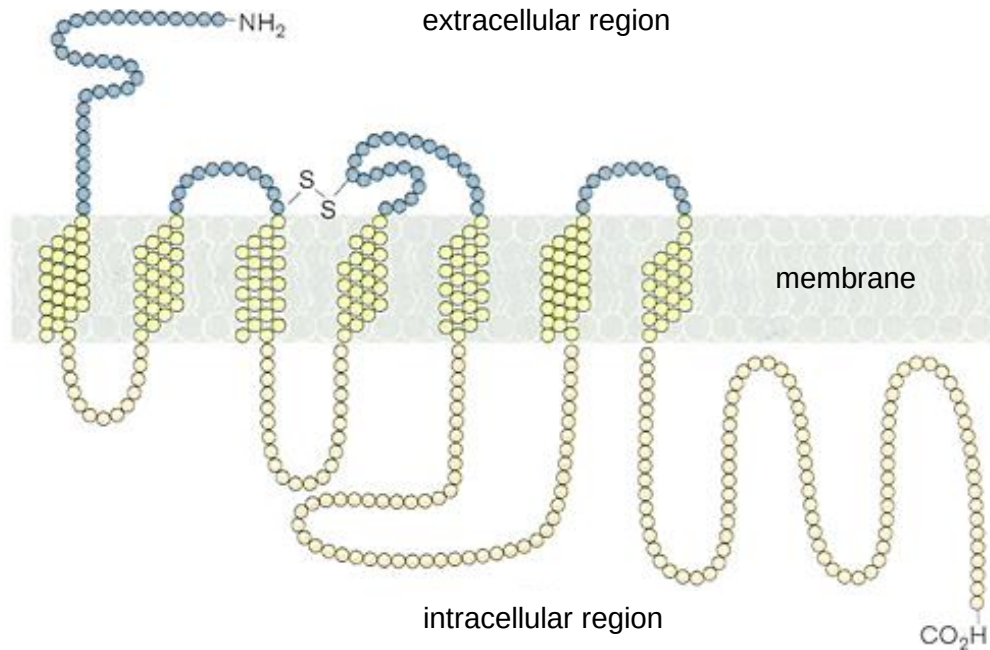


Fig.1.1

- (a) Name the type of receptor. [1]
G protein linked/coupled receptor*;
- (b) Describe the various levels of organisation in the structure of this receptor. [4]
1. The primary structure is made up of the unique sequence of the 485 amino acid residues in a polypeptide linked by peptide bonds*;
 2. The secondary structure consists of the 7 transmembrane alpha-helices;
 3. which are linked by hydrogen bonds* formed between C=O* group of one amino acid residue and N-H* group of another amino acid residue, four amino acids away of the polypeptide backbone;
 4. The tertiary structure involves single polypeptide chain further extensively folded and bent into specific 3D conformation;
 5. held in place by hydrophobic interactions*, ionic bonds*, disulfide bridges* and hydrogen bonds* formed between R groups;
- or
 the disulphide bond* between R groups* of cysteine residues; there may be others but not shown here;

- (c) A group of cell surface membrane proteins which transport sugars out of a cell have been identified and called SWEETs. They can be found in both plant and animal cells.

Each SWEET is a protein with seven coiled regions which together make a pore across a membrane bilayer shown in Fig. 1.2.

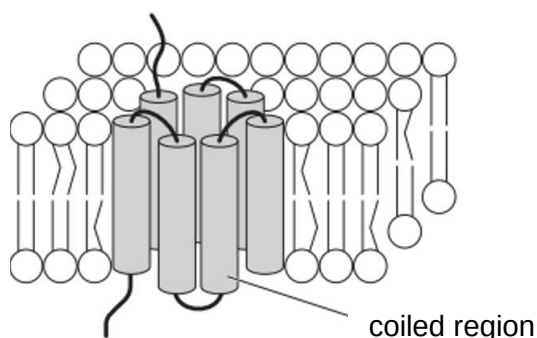


Fig. 1.2

- (i) Compare the visible structures of the transmembrane proteins shown in Fig. 1.1 and Fig. 1.2. [2]

Similarities:

- Both are made of 7 coiled regions/alpha helices that transverse through the cell membrane seven times;
- Both are made up of single polypeptide chains/ one amino and one carboxyl end;

Differences:

- The SWEET protein forms a hydrophilic pore whereas the GPCR does not;
- GPCR has a G protein binding site but SWEET protein does not;
- Coiled regions of SWEET arranged in a ring while coiled regions in GPCR arranged in a row;
- AVP : Disulphide bond / Ligand binding site in GPCR but absent in SWEET

At least 1 similarity and 1 difference

- (ii) How do the transmembrane proteins in Fig. 1.1 and 1.2 differ in their roles? [2]

Point of difference	GPCR	SWEET
1. Binding	The <u>GPCR binds to a specific ligand</u>	SWEET protein <u>does not bind to any ligands</u> but allows specific sugars through;
2. Function	The <u>GPCR allows for signal transduction</u>	<u>SWEET protein allows the transport of sugars</u>
3. Direction	<u>from the extracellular region to the intracellular region of the cell</u>	<u>from intracellular region of the cell to the extracellular region of the cell</u> ;

- (iii) Name a type of cell in an animal that has both the transmembrane proteins shown in Fig. 1.1 and 1.2. [1]

Liver cell;

[Total: 10]

- 2 Fig. 2.1 shows part of a macrophage seen under an electron microscope.

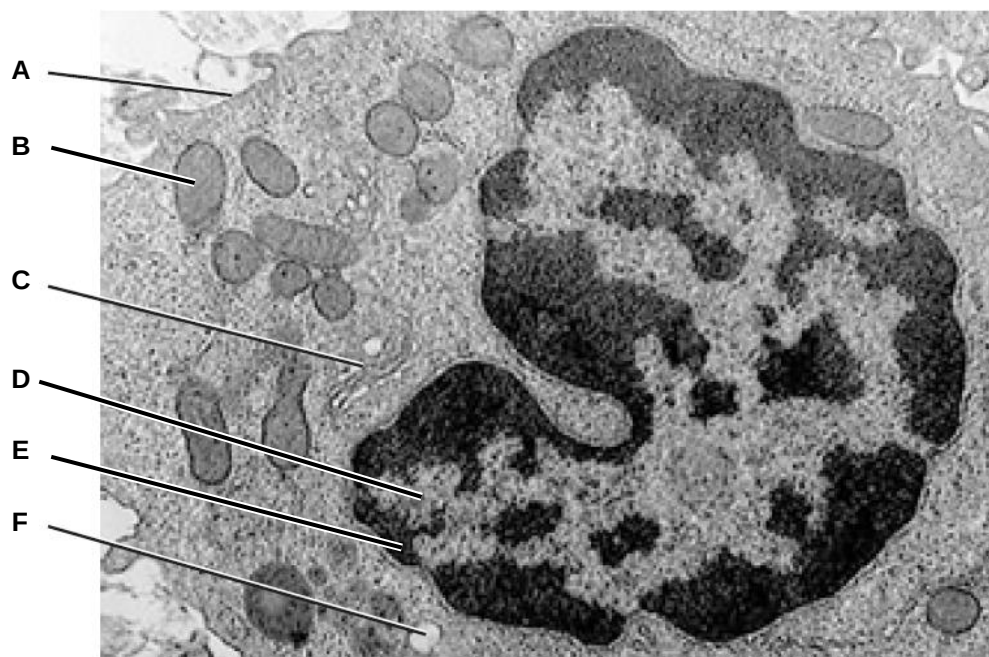


Fig. 2.1

- (a) (i) The table below gives information about the cell structures labelled in Fig. 2.1.

Complete the table by filling in the empty boxes. [2]

structure labelled	name of structure	description of structure
B	mitochondrion	1. <u>Inner membrane folded to form cristae</u> 2. <u>Contains circular DNA</u> 3. <u>Double membrane</u> 4. <u>Rod-shaped/spherical</u>
D	euchromatin	Loosely coiled DNA that is transcriptionally active.

[2]

- (ii) Explain the importance of structure A in a macrophage during phagocytosis. [2]

1. A is the cell surface/plasma membrane of the macrophage;
2. where pseudopodia* are formed and extended outwards/evagination of membrane to engulf pathogens;
3. Cell surface membrane is fluid* and able to enclose/ engulf the pathogen within a vesicle/phagosome/endosome;
A: fluidity with reference to formation of pseudopodia
4. AVP;

- (b) An experiment was carried out to study the synthesis and movement of proteins within macrophages.

Fig. 2.2 shows the method used and Table 2.1 shows the results of this experiment.

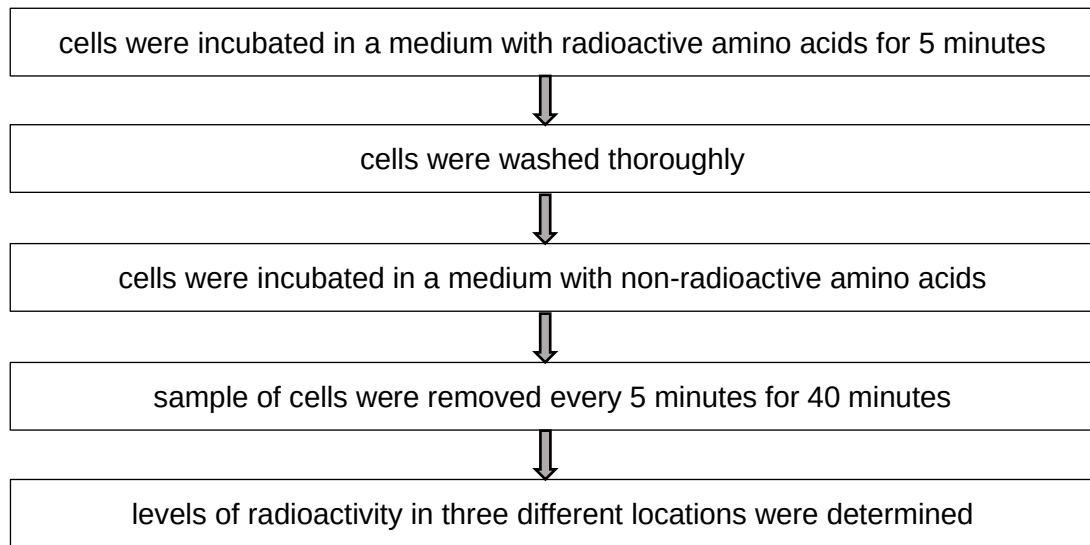


Fig. 2.2

Table 2.1

time / minutes	level of radioactivity / arbitrary units		
	ribosomes	cisternal space of rough endoplasmic reticulum	Golgi apparatus
5	60	0	0
10	80	40	0
15	20	50	0
20	15	50	10
25	10	20	30
30	5	20	60
35	5	10	40
40	5	5	20
total radioactivity	200	195	160

- (i) Describe and explain the changes in the level of radioactivity in the ribosomes. [3]
1. The level of radioactivity increased initially from 60 arbitrary units (a.u.) to 80 a.u. from 5 to 10 minutes;
 2. as ribosomes are the site of translation*/ protein synthesis where the radioactive amino acids are used as monomers to synthesise polypeptides;
 3. The level of radioactivity then decreased from 80 a.u. to 5 a. u. after 10 minutes until 30 minutes and remained low at 5 a.u. from 30 minutes to 40 minutes; A: 80 a.u. to 5 a.u from 10 to 40 mins
 4. The synthesized polypeptides have left the ribosomes and entered the cisternal space of rough endoplasmic reticulum;
A: polypeptides synthesized on free ribosomes are released to be used in the cytoplasm
- (ii) Suggest why the total radioactivity in the Golgi apparatus was lower than the total radioactivity in the ribosomes. [2]
1. some proteins are synthesised on free ribosomes for intracellular use;
 2. not all proteins need modification &/or packaging in the Golgi apparatus;
 3. some proteins may be still in transport vesicles and have yet to reach the Golgi apparatus;
 4. some proteins had left the Golgi apparatus already (e.g. MHC proteins embedded on the cell surface membrane, secreted as cytokines);

[Total: 9]

3 Fig 3.1 shows the intermediates and key stages of the Calvin cycle.

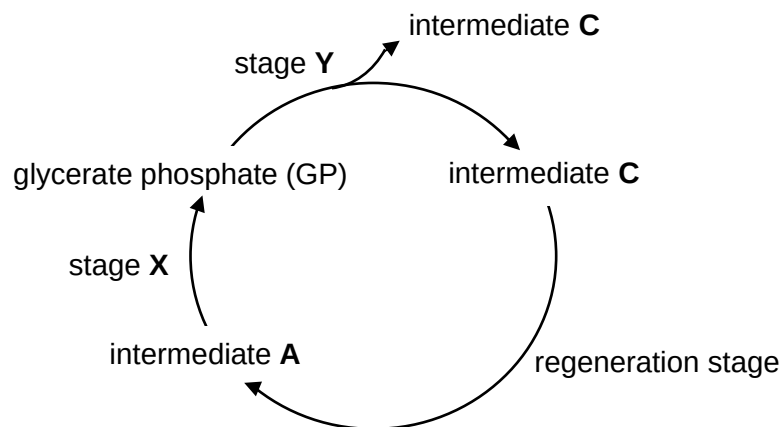


Fig 3.1

- (a) (i) Identify intermediates A and C. [1]
A: Ribulose biphosphate (RuBP)
C: triose phosphate / glyceraldehyde-3-phosphate / 3-phosphoglyceraldehyde
- (ii) Name the molecule(s) required to convert GP to intermediate C. [1]
ATP & NADPH

Fig 3.2 shows the results of an experiment. Unicellular green algal culture was grown in ample light under normal CO_2 conditions. At the time indicated in Fig. 3.2, radioactive $^{14}\text{CO}_2$ was added to the algal culture. The levels of radioactivity of intermediate A and GP were measured throughout the experiment.

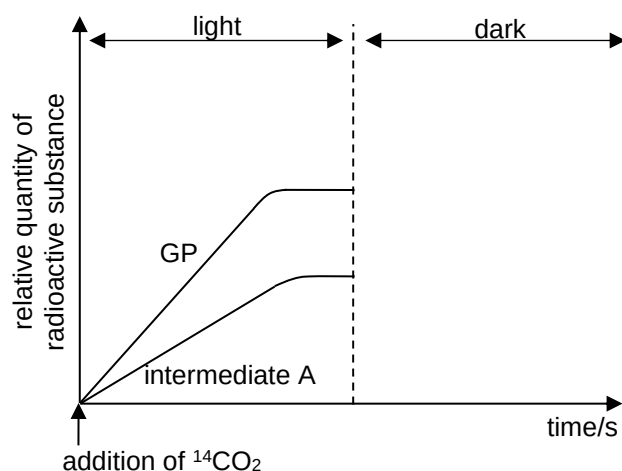


Fig. 3.2

- (b) With reference to Fig 3.2, explain the increase in the quantity of radioactive GP. [3]
1. During carbon fixation*, radioactive carbon dioxide* combines with RuBP*; A: carbon is fixed/ carboxylation;
 2. to form an unstable 6C compound that will immediately split to form 2 molecules of radioactive glycerate phosphate* (GP);
 3. Carbon fixation is catalyzed by enzyme, RuBP carboxylase/oxygenase* (Rubisco);
 4. The rate of formation of GP is greater than the rate of formation of G3P;
- (c) In the middle of the experiment, the algal culture was placed in the dark.

On Fig 3.2, extend and label the lines to predict the changes in the quantity of radioactive labeled in the following compounds after being placed in the dark :

- (i) GP [1]
Line showing increase. Accept if flat line before increase
- (ii) intermediate A [1]
Line showing decrease. Accept if flat line before decrease

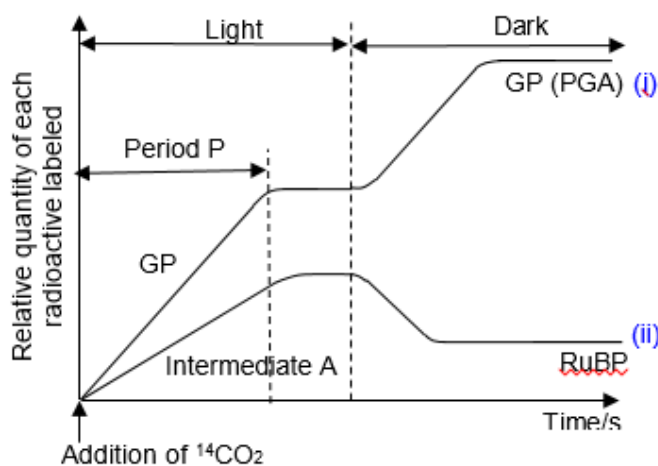


Fig. 3.2

(d) Explain your predicted trends given in part (c). [4]

1. In the dark, light-dependent reaction* does not occur and therefore ATP* and NADPH* is not produced;

Increase in GP

2. Carbon fixation* of RuBP/ RuBP is carboxylated and converted to GP as this process doesn't require ATP or NADPH;
3. GP accumulates in the dark because GP is not reduced into G3P due to lack of ATP and NADPH. Thus, the amount of GP increases;

Decrease in RuBP

4. RuBP regeneration* is slowed due to lack of ATP (and NADPH);
5. and lack of G3P for RuBP regeneration*. (mark once for RuBP regeneration);
6. RuBP is still carboxylated and converted to GP, thus results in decrease.

Must address both graphs to get full marks.

[Total: 11]

- 4 In natural environments, bacteria such as *Escherichia coli* (*E. coli*) predominantly live in communities termed biofilms, in which a self-produced extracellular matrix holds the cells together. Bacterial biofilms are found in virtually every ecosystem on Earth, from aquatic systems to terrestrial environments and the human body. Furthermore, biofilms are associated with persistent and severe infections because of their ability to colonise medical devices and implants. The biofilm structures offer the bacteria protection against hostile environments and, more worryingly, against antibiotic treatments.

The biofilm environment provides a high cell density and close cell-cell proximity that may facilitate gene transfer through bacterial conjugation.

(a) (i) Compare the processes of bacterial conjugation and transformation. [2]

	Conjugation	Transformation
Similarities		
Transfer of genetic material	Both involve introduction of new <u>genes</u> (A: DNA, alleles) from a <u>donor to a recipient</u>	
Vector	Both <u>do not require a viral vector</u> such as a phage	
Differences		
Genetic material transferred	<u>F plasmid</u> (extra-chromosomal DNA) A: F plasmid DNA R: F ⁺ plasmid / plasmid	<u>Fragments</u> of foreign naked <u>DNA</u> R: chromosomal DNA (this is too big)
Source of new genetic material	From <u>F⁺ donor</u> cell	From <u>lysed bacterial</u> cells
Amount of genetic material transferred	<u>More genes/</u> DNA transferred	<u>Fewer genes/</u> less DNA transferred
Method of transfer	Via cytoplasmic <u>mating bridge</u>	Via <u>cell surface proteins</u> of competent cells A: <u>pores</u> made by <u>heat shock</u> treatment
Recombination	Homologous recombination does not occur	Homologous recombination occurs
Contact between donor and recipient cells	Contact needed between <u>F⁺ donor</u> and <u>F⁻ recipient</u> cells R: between F ⁺ and F ⁻ cells	No contact needed between <u>donor</u> and <u>recipient</u> cells

Must be 1 similarity and 1 difference

(ii) Conjugation drives the evolution of bacteria.

Explain why this is so. [2]

1. Conjugation increases genetic variation by direct transfer of new genes/alleles from one bacterial cell to another via the F plasmid;
R: genetic diversity, variety, variance
2. natural selection/selection pressure can act on these new genes/alleles resulting in a change in allele frequency;
OR
the transfer of new alleles results directly in the change in allele frequency within the bacterial population;

(b) During conjugation in *E. coli*, the donor cell transfers an F plasmid to the recipient cell.

Fig. 4.1 shows the structure of the *tra* operon found on the F plasmid.

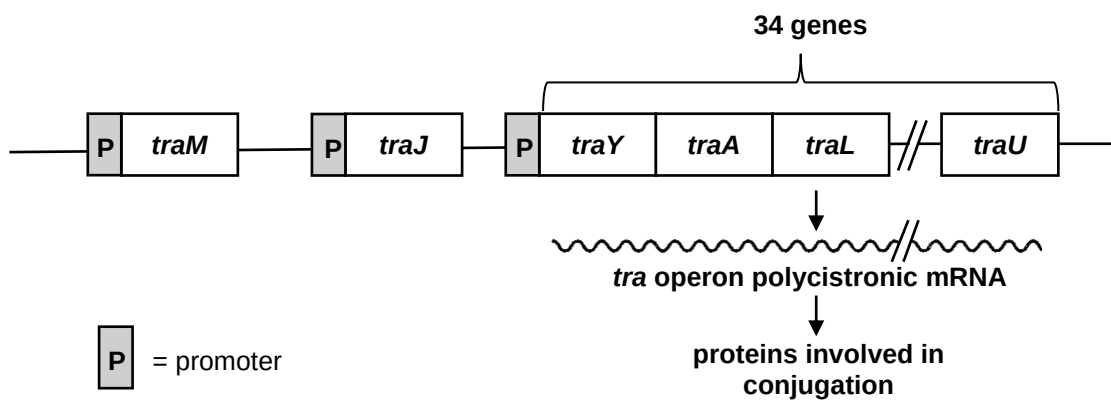


Fig. 4.1

- The majority of *tra* genes are transcribed under a single promoter as part of a large operon. There is no operator sequence in this operon.
- Regulation of the *tra* operon is controlled by the *traJ* gene which is located upstream of the operon.
- *traJ* encodes a protein that binds to the promoter of the *tra* operon to activate it.
- *traM* encodes a relaxosome protein which makes a site-specific DNA cut (nick) on one of the strands of the double-stranded plasmid at the origin of transfer.

(i) Describe the role of the *tra* operon polycistronic mRNA. [2]

1. It is the template for the **translation*** of functionally related protein products/ proteins involved in conjugation;
2. Such that the 34 structural genes are synthesised together/at the same time / from a single mRNA (A: multiple start and stop codons);
3. this allows the *E. coli* to make economical use of energy and resources/conserve resources;

- (ii) Describe the process by which the relaxosome protein facilitates conjugation. [3]
1. The relaxosome has a binding site that is complementary in shape and charge* to the origin of transfer;
 2. The relaxosome breaks the phosphodiester bond and initiates the transfer of one strand of the double stranded plasmid DNA from the donor/F⁺ cell;
 3. The 5' end of the nicked single-stranded DNA is transferred to the recipient/F⁻ cell through the mating bridge via the rolling circle mechanism (R: sex pilus);
 4. While the 3' end serves as a start point for the synthesis of the complementary strand
 5. using the other DNA strand as a template for elongation;
- or
- transferred strand in F⁻ cell used as a template for elongation;

Recent studies have found that the *tra* operon is repressed by bacterial host-nucleoid associated (H-NS) proteins which are encoded by the bacterial chromosome. Fig. 4.2 shows the location where the H-NS protein binds to on the operon.

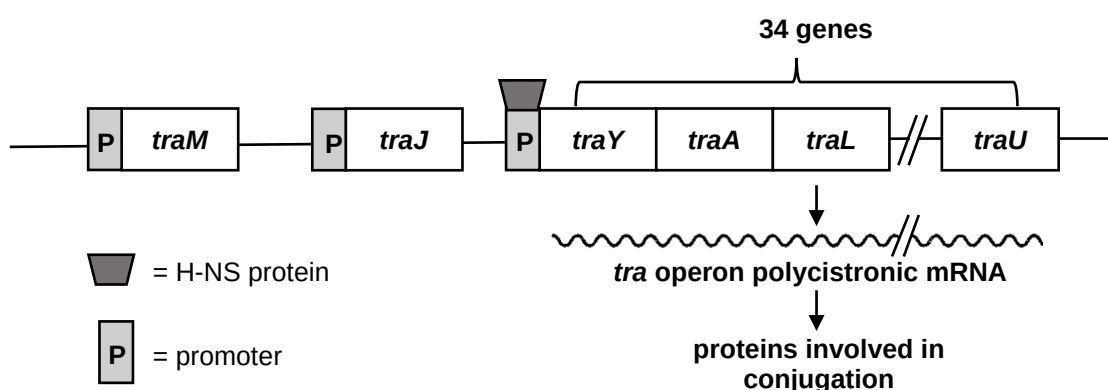


Fig 4.2

- (iii) With reference to Fig. 4.2, explain how the H-NS protein represses the *tra* operon. [2]
1. The H-NS protein has a DNA binding domain which is complementary in shape/conformation* to the promoter* of the *tra* operon;
R: operator since figure shows H-NS protein binding to promoter.
 2. When it binds to the promoter, it prevents access of RNA polymerase* from binding to the promoter and prevents transcription of the structural *tra* genes;
 3. When it binds to the promoter, it blocks Tra J protein from binding to activate the operon;

- (c) Another gene, *traT*, is located at a separate locus on the F plasmid. This gene is not required for sex pilus synthesis and DNA transfer. Instead, *traT* codes for a membrane protein.

When expressed in a bacterial cell, the TraT membrane protein prevents a sex pilus from another F⁺ cell from binding to it.

Suggest why it is advantageous for the F plasmid to contain the *traT* gene. [1]

1. Limits ability of plasmid carrying cells to serve as a recipient for the same plasmid again (idea of 2 plasmids);
2. Preventing self-mating can avoid metabolic cost and potential death associated with repeated plasmid transfer
3. AVP

[Total: 12]

- 5 In tomato plants, stem colour, leaf shape and leaf colour are controlled by three different genes that are found on the same chromosome. The allele for red stems, **R**, is dominant to the allele for green stems, **r**. The allele for tomato leaf shape, **T**, is dominant to the allele for potato leaf shape, **t** and the allele for dark green leaves, **D** is dominant to the allele for light green leaves, **d**. The two tomato plants described below were crossed together.

Parent 1
red stem
tomato leaf shape
dark green leaves

Parent 2
green stem
potato leaf shape
light green leaves

- (a) The parent plants were both pure-breeding.

State the genotype and phenotype of the F_1 plants. [1]

Genotype: $RrTtDd$ (A: stick diagram)

Phenotype: Red stem, tomato leaf shape and dark green leaves.

need both correct to get 1 mark

- (b) The F_1 plants were then selfed to produce an F_2 generation of plants. Since the genes are linked, many of the offspring in the F_2 generation were expected to show mostly the two original parental phenotypes.

Assuming there is no crossing over between the linked genes,

- (i) state the genotypes of the F_2 plants showing the same phenotypic characteristics as parent 1; [1]

$RRTTDD$ and $RrTtDd$;

- (ii) give the expected genotypic ratio of all the F_2 offspring. [1]

$RRTTDD$: 2 $RrTtDd$: 1 $rrttdd$

In fact, there were eight different phenotypes among the F_2 plants. This can be explained by crossing over having taken place.

Fig. 5.1. represents the behaviour of one pair of chromosomes during meiosis.

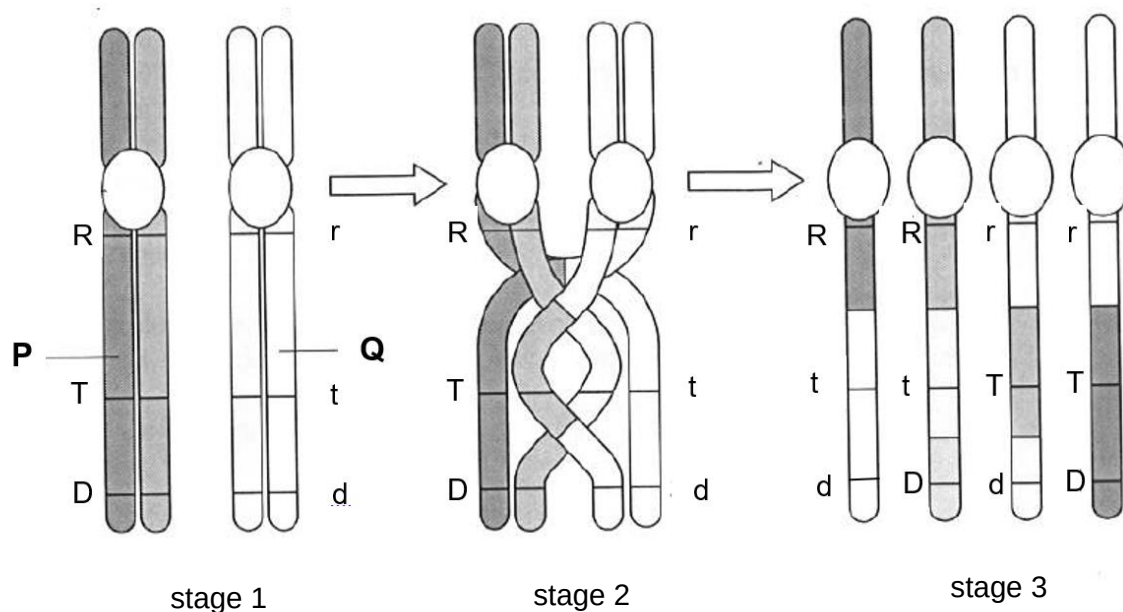


Fig. 5.1

(c) With reference to Fig. 5.1, compare structure **P** with structure **Q**. [3]

1. **P & Q are non-sister chromatids* of homologous chromosomes;**

Similarity

2. They have the same sequence of **genes*** (e.g. genes coding for stem colour, leaf shape and leaf colour) at the same loci;

3. They are also joined to their own identical sister chromatid at the same centromere position.

4. same size/ length

Differences

5. **P carries dominant allele D while Q carries recessive allele d** (cite any 2 alleles)

(A: Structures P & Q carry different alleles at the same gene loci (e.g.); WTTE

6. This is due to each homologue being inherited from each of the parents (i.e. maternal and paternal chromosomes);

Must have one similarity and one difference to get full marks

(d) Describe the events that occurred in stage 2 in Fig. 5.1. [2]

1. **Prophase I*** of meiosis where synapsis occur when homologous chromosomes pair up;

2. **chiasma*** formed and **crossing over*** occur between **non-sister chromatids*** of the homologous pair;

3. by the breaking and rejoining of exchange of equivalent portions of the chromatids;

(e) A farmer collected two tomato plants, T1 and T2, which were produced from crosses between different pure-breeding plants in his garden. Both have dark green leaves and red stems. He conducted a test cross on each plant. Table 5.1 shows the results of these crosses.

Table 5.1

phenotype	observed numbers	
	plant T1	plant T2
dark green leaves and green stems	122	2
light green leaves and green stems	6	82
dark green leaves and red stems	4	85
light green leaves and red stems	124	4

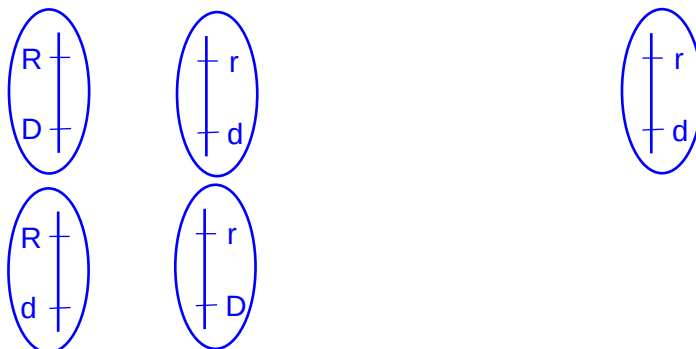
- (i) Draw a genetic diagram to explain the results of the test cross for plant T2. [3]

Parental phenotype Dark green leaves and red stems (plant T2) x light green leaves and green stems

Parental genotype



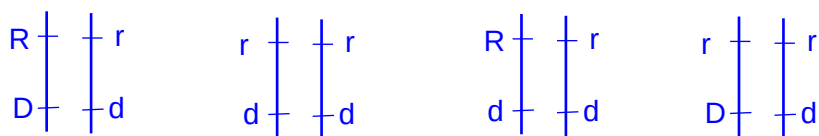
Gametes



Offspring phenotype and genotype:

Gametes	$\begin{array}{c} R \mid \\ D \mid \end{array}$	$\begin{array}{c} \mid r \\ \mid d \end{array}$	$\begin{array}{c} R \mid \\ d \mid \end{array}$	$\begin{array}{c} \mid r \\ \mid D \end{array}$
$\begin{array}{c} \mid r \\ \mid d \end{array}$	$\begin{array}{c} R \mid \mid r \\ D \mid \mid d \end{array}$	$\begin{array}{c} r \mid \mid r \\ d \mid \mid d \end{array}$	$\begin{array}{c} R \mid \mid r \\ d \mid \mid d \end{array}$	$\begin{array}{c} r \mid \mid r \\ D \mid \mid d \end{array}$
	Dark green leaves and red stems	Light green leaves and green stems	Light green leaves and red stems	Dark green leaves and green stems

Offspring genotype:



Offspring phenotype:

Dark green leaves and red stems Light green leaves and green stems Light green leaves and red stems Dark green leaves and green stems

Parental phenotypes
Recombinant phenotypes

Phenotypic frequency:

85 82 2 4

1. Correct parental genotype and phenotype – drawn with stick diagrams;
2. Correct gametes and circled;
3. Correct offspring genotype and corresponding phenotype;
4. Minus 1 mark for not doing stick diagram

- (ii) Account for the different test cross results obtained from plants T1 and T2. [3]
1. Both plants T1 and T2 have different allelic arrangements on their chromosomes/ produces gametes with different allelic combinations;
 2. because in plant T1, alleles R and d are on the same chromosome while alleles r and D are on the same chromosome;
 3. while in plant T2, alleles R and D are on the same chromosome while alleles r and d are on the same chromosome;
 4. For T1's test cross, more parental phenotypes (dark green leaves and green stems, and light green leaves and red stems) and fewer recombinant phenotypes (light green leaves and green stems and dark green leaves and red stems) because crossing over is a chance event (A: for T2);

[Total: 14]

- 6 Telomere length is maintained in most eukaryotic cells by telomerase. The core components of this ribonucleoprotein (RNP) complex include a protein catalytic subunit (TERT), and an RNA subunit that contains a short template sequence essential for the synthesis of telomeric repeats.

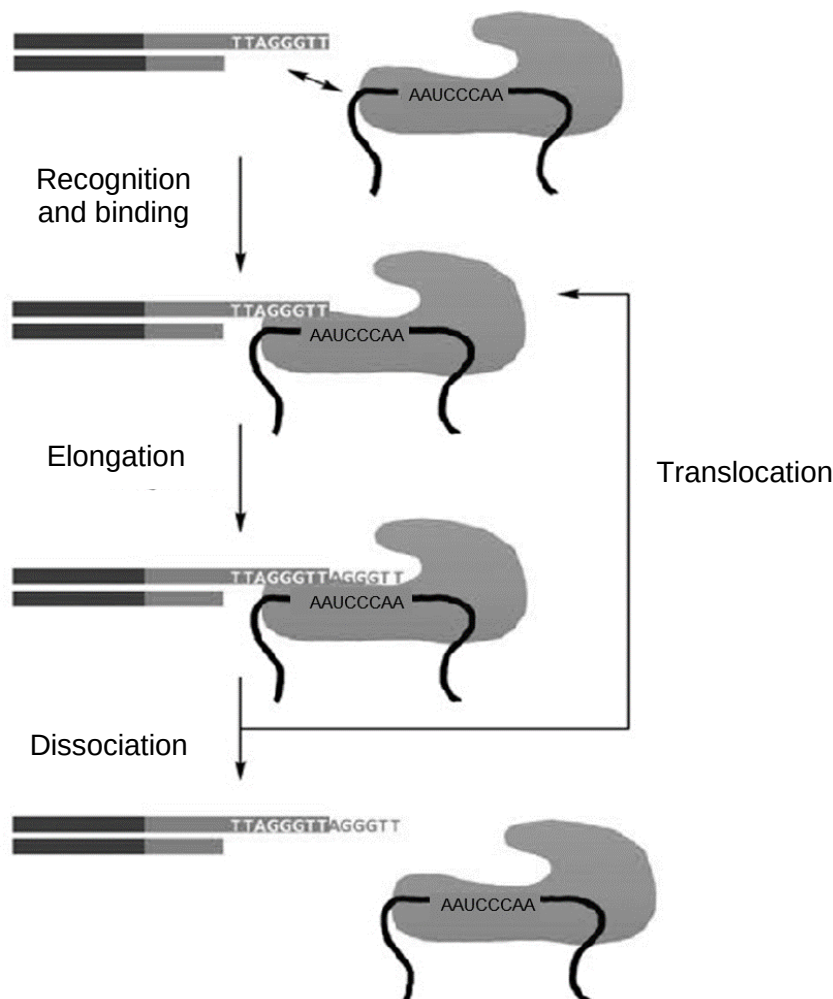


Fig. 6.1

- (a) (i) With reference to Fig. 6.1, explain the role of the TERT protein in telomerase. [3]
1. TERT has an **active site*** that is **complementary in conformation and charge*** to a **specific telomeric DNA sequence**;
 2. Using **telomerase RNA** as a **template***, telomerase reverse transcriptase forms a complementary **DNA*** sequence through **complementary base pairing***;
 3. Catalyzes the **formation of phosphodiester bonds*** between deoxyribonucleotides,
 4. Translocation in the **5' → 3' direction** to produce a series of **tandem/ telomeric repeats** thus **elongating the telomere**;

- (ii) Telomerase activity is observed to be high in stem cells but downregulated in differentiated cells.

Explain why high telomerase activity is needed in stem cells but not in differentiated cells. [3]

1. To maintain a constant pool of stem cells through self-renewal*/continue to differentiate* into specialised cells.
2. Each round of DNA replication is accompanied by shortening of telomeres* at the ends of the chromosomes; (A: end replication problem)
3. When telomeres are shortened to a critical length, (A: reach Hayflick limit) cells stop dividing / divide a limited number of times /undergo programme cell death/ triggered pathway to apoptosis/ preventing loss of vital genetic information
4. Stem cells undergo continuous proliferation while differentiated cells do not;
5. Stem cells but not differentiated cells therefore express telomerase gene to produce telomerase* that lengthens telomeres thus allowing them to proliferate continuously/avoid undergoing apoptosis.

As with stem cells, cancer cells are also known to exhibit high levels of telomerase activity.

The high levels of telomerase activity in cancer cells may be contributed by mutations which involve the *TERT* locus, as shown in Fig. 6.2 below.

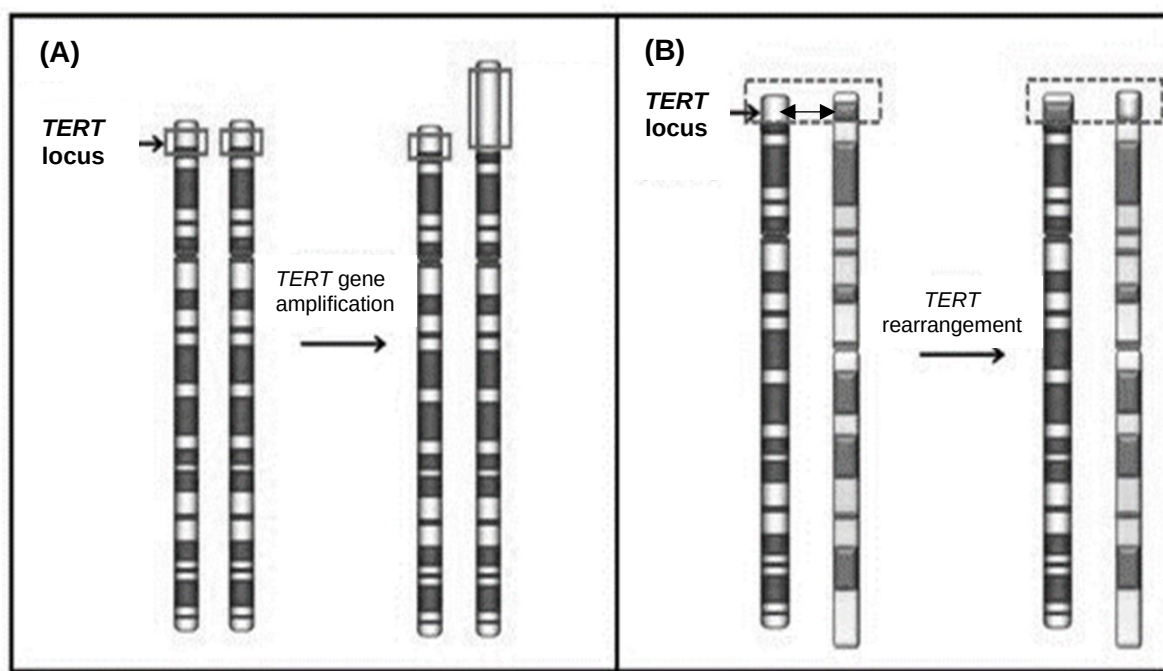


Fig. 6.2

- (b) Suggest how the mutations that involve the *TERT* locus shown in the Fig. 6.2(A) and Fig. 6.2(B) could result in the high levels of telomerase activity in cancerous cells.

- (i) Fig. 6.2(A): [1]

Gene amplification resulting in more copies of the *TERT* gene, therefore the expression of more TERT proteins and increased telomerase activity.

- (ii) Fig. 6.2(B): [1]

1. Translocation resulting in the *TERT* gene falling under the control of an enhancer and increasing the expression of TERT proteins and telomerase activity.

R: insertional inactivation of any gene since this is not supported by the banding in the figure.

- (c) In the laboratory, differentiated cells can be made into induced pluripotent stem cells (iPSCs). This is done by introducing transcription factors into the differentiated cells.
- (i) Explain how transcription factors work in the production of iPSCs from differentiated adult cells. [2]
- Transcription factors may act as repressors* which bind to silencers; or
Transcription factors may act as activators* which bind to enhancers;
 - Upregulate/downregulate the transcription of genes in the differentiated somatic adult cells; WTTE
 - and alter the gene expression patterns of the cells such that it is similar to that of embryonic stem cells; resulting in the production of iPSCs;
- (ii) Explain how iPSCs overcome an ethical concern of using embryonic stem cells in therapy. [1]
- Since iPSCs can be derived directly from adult tissues, its production does not generate or destroy any human embryos/ does not treat the embryo as a source of spare parts;

[Total: 11]

- 7 Fig. 7.1 shows the sequence of events in the insulin signaling pathway in a muscle cell.

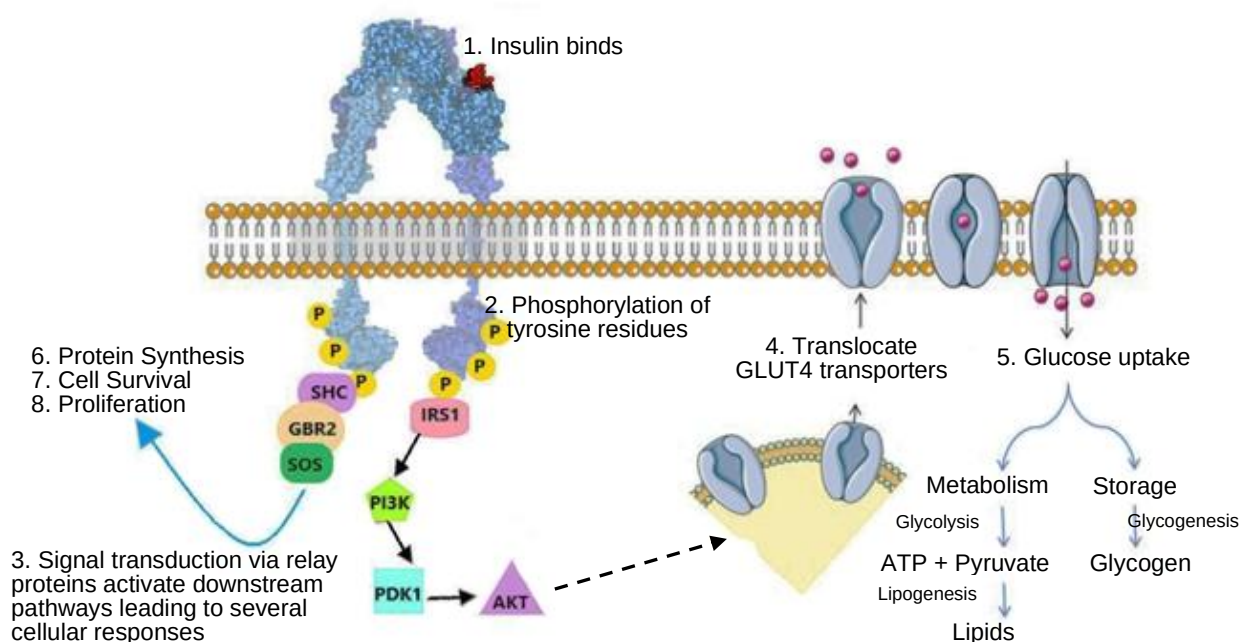


Fig. 7.1

With reference to Fig. 7.1,

- (a) (i) describe how phosphorylation of tyrosine residues occur; [3]
- Insulin binds to extracellular ligand-binding site of specific transmembrane receptor by complementary shape* and charge;
 - Conformational change in the intracellular domain of receptor results in activation of intrinsic tyrosine kinase;
 - Tyrosine kinase has an active site* complementary in shape and charge to ATP and tyrosine residues;
 - Intrinsic kinase activity of each subunit in the intracellular domain cross-phosphorylates /autophosphorylates the 6 tyrosine residues /Addition of 6 phosphate groups to the 6 tyrosine residues;

(ii) suggest how activation of AKT can lead to an increase in glucose permeability of muscle cells. [2]

1. With the activation of AKT, vesicles move towards cells surface membrane along microtubules;
2. these act as tracks along which vesicles containing GLUT 4 glucose transporters translocate/move towards and fuse* with the plasma membrane; (Reject: diffuse)
3. This increases the number of GLUT 4 glucose transporters within the plasma membrane;
4. allowing more glucose to enter the muscle cells via facilitated diffusion, thus increasing glucose permeability of muscle cells;

(b) AKT will also activate an enzyme, glycogen synthase, which will catalyse the conversion of glucose into glycogen.

Explain how the structure of glycogen relates to its role as a storage molecule. [2]

1. Large molecule/long chain so insoluble in water ; (or A: Intramolecular H bonding, fewer OH groups available to interact with water)
2. Hence does not affect water potential of cells / no osmotic effect ;
- or
3. Helical molecule so more glucose* units per unit volume,
4. thus compact structure for storage ;
- or
5. Extensively branched so provides many ends for hydrolysis of glycosidic bonds ;
6. which releases more glucose* molecules per unit time ;

(c) Another cellular response to insulin apart from glycogen synthesis is the increased rate of respiration of the muscle cells using glucose as a respiratory substrate.

Fig. 7.2 represents two stages of aerobic respiration that take place in a mitochondrion.

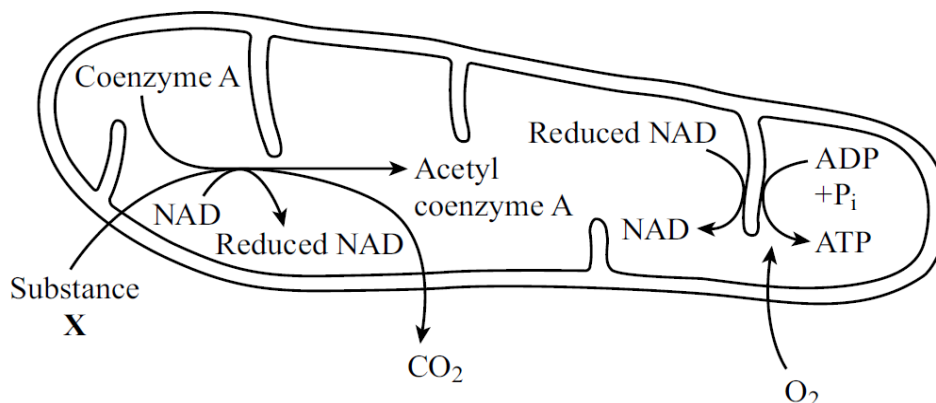


Fig. 7.2

(i) Name substance X. [1]
Pyruvate;

(ii) Which stage of aerobic respiration takes place inside a mitochondrion but is not represented on the diagram? [1]
Krebs Cycle

(iii) Explain why oxygen is needed for the production of ATP on the cristae of the mitochondrion. [3]

1. Oxygen is needed as the final electron acceptor* of the electron transport chain (ETC);
2. so that NADH and FADH can continue to donate electrons to the ETC;
3. As electrons are passed down the ETC, energy released is used to pump protons from the matrix to the intermembrane space;
4. Produces a proton motive force which allows protons to diffuse through ATP synthase*;
5. This diffusion of protons is coupled to the phosphorylation of ADP to form ATP* via chemiosmosis*;

All human cell surface membranes and some internal membranes contain transmembrane glucose transporters (GLUTs).

There are at least 14 different GLUTs, each with differing functions. For example, GLUT1 ensures that sufficient glucose enters cells to supply their energy requirements.

(d) Mature red blood cells do not contain mitochondria and have relatively low energy requirements. Despite this, the membranes of red blood cells contain more GLUT1 molecules than other cells with similar energy requirements.

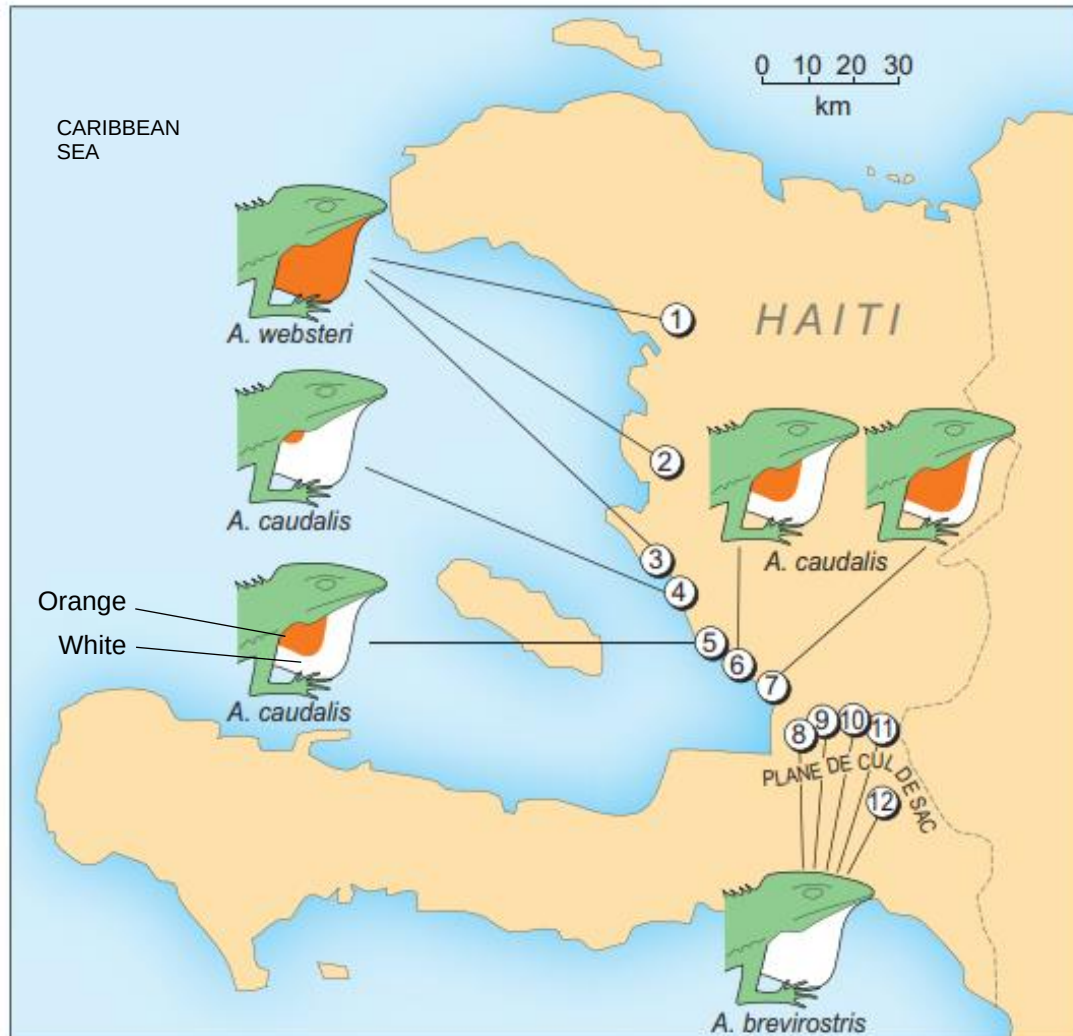
Explain why the cell surface membranes of red blood cells require large numbers of GLUT1 molecules to supply their energy requirements. [2]

1. Absence of mitochondria, ATP required by red blood cells is produced only through glycolysis*;
2. Hence, only 2 nett ATP will be produced per glucose molecule;
3. For the same quantity of ATP required by normal cells, red blood cells will need 18x more glucose
or
red blood cells produce 36 less ATP molecules per glucose molecule, than normal cells, so need more glucose
4. hence more GLUT1 molecules to transport glucose into the red blood cells;
hence more GLUT1 molecules needed to transport glucose into the RBCs.

[Total: 14]

- 8 In the Caribbean islands, there is a group of lizards called anoles. The males possess a larger retractable flap of skin under the throat, termed a dewlap, which is used in courtship, aggressive interactions and even as a warning when encountering predators.

The patterns of orange colouration on the dewlaps of 3 species of the anole lizard and the distribution of 12 population study sites in Haiti are shown below.



zone	population study sites	habitat
north	1, 2, 3	beach
central	4, 5, 6, 7	beach
south	8, 9, 10, 11, 12	forest

Fig. 8.1

- (a) Define the term species. [2]

1. A group of organisms of the same species are capable of interbreeding* and producing fertile, viable offspring*;
2. are reproductively isolated* from other species;
3. They have a common gene pool and the same chromosome number;
4. They usually have similar morphological, physiological and behavioural features;
5. AVP
(other species concepts, e.g. ecological/morphological species concept, correctly and fully defined)

(b) The patterns of orange and white dewlap colouration for each anole species is different.

(i) Explain how the habitat in the south zone could bring about the dewlap colouration found in *A. brevirostris*. [4]

1. Anoles/lizards with white dewlaps were more visible in the dim forest habitat and had a selective advantage;
2. ref. to one relevant reason
reason: as males with white dewlaps were able to attract more females to mate;
or
reason: as anoles with white dewlaps were able to scare off predators easily;
3. Hence individuals with white dewlaps were selected for/ survived and reproduced, and passed on their alleles coding for white dewlaps to the next generation;
4. Thus favourable white dewlap allele frequency increased (R: changes);
which eventually lead to all the anoles in the south having white dewlaps over time

(ii) Breeding experiments in the lab have revealed that dewlap colour-pattern follows Mendel's laws of inheritance, in which orange (**D**) is dominant to white (**d**).

Although having a white colouration is disadvantageous at the beach habitats, the frequency of allele **d** in *A. websteri* has not reached zero even after many generations.

Explain why this is so. [2]

1. The expression of recessive allele d is masked by the expression of the dominant allele D in heterozygotes with orange dewlaps;
2. Since natural selection only acts on phenotypes and not genotypes;
3. the frequency of allele **d** will never be zero as the d allele heterozygotes with orange dewlaps will be passed on the offspring;
4. This is known as heterozygote protection* and it occurs in diploid organisms;

(c) Suggest how anole lizards could have moved from Haiti to another island. [1]

1. They may have floated on natural rafts;
2. swam;
3. AVP;

[Total: 9]

- 9 The treatment for people with active tuberculosis (TB) lasts 6 months and involves a combination of antibiotics. This is usually very effective if the person has a susceptible (non-resistant) strain of *Mycobacterium tuberculosis*, the causative organism of TB.

Table 9.1 summarises one recommended treatment strategy that involves a combination of antibiotics.

Table 9.1

antibiotic	length of treatment	mode of action of antibiotic
rifampicin	6 months	enters bacterial cells and inhibits RNA polymerase activity
isoniazid	6 months	prevents the synthesis of cell wall components known as mycolic acids
ethambutol	first 2 months	prevents mycolic acids from being added to the cell wall
pyrazinamide	first 2 months	prevents the synthesis of fatty acids

- (a) Susceptible strains of *M. tuberculosis* will be killed using any one of the antibiotics listed in Table 9.1. However, combination treatment is preferred to reduce the prevalence of antibiotic resistance.

With reference to Table 9.1, explain how combination treatment for TB can help to reduce the prevalence of antibiotic resistance compared to single antibiotic treatment. [2]

1. Combination treatment of antibiotics, act on/target different parts of the bacterial replication cycle / have different modes of action;
2. Bacteria are likely to gain resistance, due to *mutations**, to a single antibiotic and are unlikely to gain resistance against all the antibiotics;
3. Hence some of the antibiotics should still be effective and all bacteria will be killed leaving no bacteria behind to develop resistance;
So prevalence of antibiotic resistance can be reduced.

- (b) Rifampicin binds tightly to an RNA polymerase molecule close to its active site. This affects the activity of the enzyme.

During the formation of mRNA, a number of events occur that involve the action of RNA polymerase.

Suggest how rifampicin can affect the activity of RNA polymerase. [3]

1. Rifampicin binds to a site close to the active site and alters shape of **active site*** by acting as a non-competitive inhibitor;
R: competitive inhibitor
A: partially blocks **active site***
2. preventing ribonucleotides from binding to /entering **active site*** of RNA polymerase;
R: substrates/ nucleotides
3. Hence RNA polymerase cannot attach to **promoter*** sequence and so;
4. unzipping / unwinding and separation of DNA double helix prevented;
(A: prevented breaking of H bonds)
5. **complementary base pairing*** cannot occur between DNA and RNA nucleotides;
6. and **phosphodiester bond*** formation between adjacent ribonucleotides is prevented;

Hence mRNA/tRNA /rRNA formation is prevented

I: stops transcription

- (c) RNA polymerase is composed of 5 different polypeptides. Gene *rpoB* codes for one of these polypeptides known as the β -subunit.

One or more mutations in a specific region of *rpoB* result in strains of *M. tuberculosis* that are resistant to rifampicin. In these strains, mutations often occur in 2 DNA triplets in this region, at positions 526 and 531.

Table 9.2 summarises the results of an investigation into 7 rifampicin-resistant strains, **A** to **G**, that have amino acid changes for positions 526 and 531.

Table 9.2 includes:

- the change in the **mRNA codon** for position 526 or position 531
- the amino acid change that has occurred as a result of the mutation
- the minimum concentration of rifampicin required to inhibit growth of the bacterial strain (MIC)
- the number of **other** mutations occurring within the specific region of *rpoB*.

Table 9.2

Key: $\approx$ approximately $\geq$ greater than or equal to $\leq$ less than or equal to

strain	codon involved	mRNA codon change	amino acid change	MIC/ μgcm^{-3}	number of other mutations in the specific region
A	526	CAC $\rightarrow$ UAC	His $\rightarrow$ Tyr	≤ 50	0
B	526	CAC $\rightarrow$ AAC	His $\rightarrow$ Asn	≥ 100	1
C	526	CAC $\rightarrow$ CGC	His $\rightarrow$ Arg	$\approx 50-75$	2
D	526	CAC $\rightarrow$ CGC	His $\rightarrow$ Arg	≥ 100	3
E	526	CAC $\rightarrow$ CGC	His $\rightarrow$ Arg	≈ 50	3
F	526	CAC $\rightarrow$ UUC	His $\rightarrow$	≥ 100	3
	531	UCG $\rightarrow$ UUG	Ser $\rightarrow$ Leu		
G	526	CAC $\rightarrow$ UAC	His $\rightarrow$	≥ 100	3
	531	UCG $\rightarrow$ UUC	Ser $\rightarrow$ Phe		

- (i) Complete Table 9.2 to show the amino acid changes that have occurred in strains **F** and **G**.
[1]
F: His $\rightarrow$ Phe
G: His $\rightarrow$ Tyr;
one mark for both correct answers

- (ii) With reference to Table 9.2, list the strains of *M. tuberculosis* that show the greatest resistance to rifampicin. [1]

B D F G;

in any order, but one mark for all 4 correct answers

- (c) Suggest reasons to explain why strains **C**, **D** and **E** show:

- resistance to rifampicin
- different levels of resistances to rifampicin. [3]

Strains C, D and E have resistance to rifampicin (max 2 marks)

1. Resistance to rifampicin occurs when there is a change in the nucleotide sequence/thymine to cytosine which results in a change in the amino acid sequence/his to arg;
2. This will change the interactions/bonds between the R groups of amino acids;
3. Which will change the tertiary (quaternary) structure of β -subunit/RNA polymerase;
4. Prevents strong binding of rifampicin to β -subunit/RNA polymerase;
Or
Lower the affinity of rifampicin to β -subunit/RNA polymerase;

Reject: if context is about rifampicin binding to the active site of RNA polymerase

Strains C, D and E have different levels of resistance to rifampicin (max 2 marks)

5. Varying number of other mutations in the specific region/ C has different number of other mutations from D and E/ C has two other mutations while D and E have three;
6. Leads to varying extent of change in the tertiary (quaternary) structure of β -subunit/RNA polymerase;
7. Resulting in different levels of resistance to rifampicin due to different affinity of rifampicin to β -subunit/RNA polymerase;
Or
Resulting in higher levels of resistance to rifampicin due to lowered affinity of rifampicin to β -subunit/RNA polymerase;
Or
Resulting in lower levels of resistance to rifampicin due to rifampicin being able to bind more strongly to β -subunit/RNA polymerase;

[Total:10]

End of Paper